

		Robert & Casella (2004) and Rizzo (2019)		
	Sections from References:			
V	Open Ended Module:		12	
	1	Generate random numbers using statistical software for different distributions with its estimation and model fitting. Apply resampling methods for real life data.		
	Sections from References:			
Books and References:				
1. Rizzo, M. L. (2019). Statistical Computing with R, second edition. Boca Raton, FL: Chapman & Hall/CRC Press				
References				
2. McLachlan, G.J. and Krishnan, T. (1997): The EM Algorithms and Extensions, Wiley.				
3. Robert, C.P. & Casella, G. (2004) Monte Carlo Statistical Methods, 2ndEdn., Springer.				
4. Ross, S. M. (2022). Simulation. Academic Press.				
5. Rubinstein, R.Y. (2017). Simulation and the Monte Carlo Methods, Wiley.				
6. Shao, J., & Tu, D. (2012). The jackknife and bootstrap. Springer Science & Business Media.				

Programme	B. Sc. Statistics				
Course Code	STA6EJ302				
Course Title	Reliability Theory				
Type of Course	Major Elective				
Semester	VI				
Academic Level	300-399				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites					
Course Summary	Determine the reliability of systems based on defined/determined reliability of the system elements and defined block diagram for the				

Objective	reliability of the observed system.
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Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Describe the structural properties of coherent systems.	U	C	Instructor-created exams / Quiz
CO2	Determine the reliability of a system.	Ap	P	Practical Assignment / Observation of Practical Skills
CO3	Discuss the different parametric distributions in reliability	Ap	P	Seminar Presentation / Group Tutorial Work
CO4	Discuss the lifetime of a system based on ageing properties	U	C	Instructor-created exams / Home Assignments
CO5	Discuss different censoring schemes.	Ap	P	One Minute Reflection Writing assignments
CO6		Ap	P	Viva Voce
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (48 +12)	Marks (70)
			10	15
	1	System of components		
	2	series and parallel structure with examples		
	3	dual structure function		
	4	coherent structure		

I	5	preservation of coherent system in terms of paths and cuts		
	6	representation of bridge structure		
	7	relative importance of components		
	8	modules of coherent systems		
	Sections from References:			
II			10	15
	9	Reliability of a system of independent components		
	10	Some basic properties of system reliability		
	11	Computing exact system reliability		
	12	Inclusion exclusion method		
	13	Reliability importance of components		
	Sections from References:			
III			16	20
	14	Reliability function, hazard function,		
	15	Residual life time, mean residual life function, one-one correspondence of these functions.		
	16	Common life distributions, exponential, weibull, gamma, pareto, lognormal and their characteristics.		
	17	Type –I, Type-II and random censoring schemes.		
	18	Likelihood functions based on these sampling schemes.		
	Sections from References:			
IV			12	20
	19	IFR, IFRA, DMRL, NBU, NBUE classes and their duals.		
	20	Exponential distribution and its aging property		
	21	Aging properties of common life distributions		
	22	Classes under formation of coherent structures.		
	Sections from References:			
V	Open Ended Module:		12	
	1	Estimation and testing based on these schemes for various		